

Introduction to Statistical Machine Learning

PSTAT 131/231

Spring 2026

INSTRUCTOR:	Dr. Tony Coburn (tony_w_coburn@ucsb.edu)
TEACHING ASSISTANTS:	Tadeas Nanjo (tnanjo@ucsb.edu) Siyu Chen (siyu_chen@ucsb.edu) Zifeng (Robin) Zhan (zifeng_zhan@ucsb.edu)
LECTURE:	TR 5:00 - 6:15 pm TD-W 1701
SECTIONS:	W 12:00 - 12:50 pm PHELP 1513 NANJO W 1:00 - 1:50 pm PHELP 1513 ZHAN W 2:00 - 2:50 pm PHELP 1513 CHEN W 3:00 - 3:50 pm PHELP 1513 CHEN W 4:00 - 4:50 pm PHELP 1513 NANJO

Course information

Description

STATISTICAL machine learning is used to discover patterns and relationships in large data sets. Topics will include: data exploration, missing data handling, linear and logistic models, regularization, classification and regression trees, random forests, gradient-boosted trees, clustering and association rules, principal components analysis, and support vector machines. Students will be guided through the process of building predictive models while focusing on model selection, model comparison, and performance evaluation. Emphasis will be on concepts, methods, and data analysis. Students are expected to complete a significant class project using real-world data. Prerequisites: PSTAT 120A-B and PSTAT 126 with a minimum grade of C or better. Credit units: 4.

THIS COURSE is taught at two levels, one aimed at undergraduates (131 level) and one at graduate students (231 level). Lectures are given at a level designed to be accessible and relevant to all students in the course. Students taking the course at the 231 level will be assigned **additional homework questions** and must do the reading in **both** the 131 and 231-level textbooks, listed below. The final project assignment is structurally the same for both levels. However, students enrolled in PSTAT 231 will be held to higher expectations with regard to mathematical depth, rigor of explanation, and interpretation of results for both homework and the final project.

Learning Outcomes

Upon completion of this course, students should be able to:

1. EXPLAIN the common statistical learning techniques conceptually and characterize some of them mathematically;
2. DESCRIBE AND DISCUSS the pros and cons of the common statistical learning methods;
3. USE R, the *tidyverse*, and *tidymodels* effectively for exploratory data analysis, model fitting, and visualization;
4. EVALUATE AND COMPARE models using appropriate resampling, performance metrics, and diagnostics;

5. CLEARLY COMMUNICATE modeling choices, limitations, and uncertainty in written analyses.

Attendance

LECTURES and sections will be conducted in person. You are strongly encouraged to attend **all** lectures and sections; failure to attend will result in your missing the content that was presented or discussed. If you do not attend a lecture or section, you are still expected to complete all assignments and readings on time.

If you are unable to attend for any **university-sanctioned reason**, or if you are ill, please inform the instructor and your TA(s) as soon as possible. If you are ill, please do not attend lecture or section.

Format

LECTURE slides will be posted on Canvas before the corresponding class. The class will progress generally according to a weekly schedule (provided below). Both lecture periods each week will consist of slides that cover related concepts, sometimes including live coding to demonstrate how to apply the concepts using the R-based language.

Section each week will be dedicated to completing an R-based lab. Section attendance is required and is **very strongly** recommended, since doing so will enable you to learn and practice your R skills in an environment where you can easily obtain help from the TA(s) and your peers. Lecture attendance, although not graded, is assumed in the grading of homework and especially of your final project.

Should you need to switch sections, please ask the TA of the section you want to switch to if there is room for you to attend. If the TA allows, you can then begin attending the new section while remaining officially enrolled in your previous section. **Sections will be held beginning Wednesday, April 8th.** Note that switching sections does not change grading expectations or attendance deadlines.

Required Materials

THE BOOKS and software used for this class are as follows. Note that all are freely available online and do not require purchase.

- **Required for PSTAT 131 students:** *An Introduction to Statistical Learning with Applications in R* (or ISLR), by G. James, D. Witten, T. Hastie, R. Tibshirani. Available: <https://www.statlearning.com/>
- **Required for PSTAT 231 students:** *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (or ESL), by T. Hastie, R. Tibshirani, J. Friedman. Available: <https://hastie.su.domains/ElemStatLearn/>
- **Required for all:**
 - The R statistical environment version 4.4.x or later, available here: <https://www.r-project.org/>
 - The RStudio IDE (integrated development environment) Desktop version. Available: <https://www.rstudio.com/>

Supplemental Materials

THE FOLLOWING are materials that are **not required** for this course, but are **recommended**. Don't feel expected to read all of these; I encourage you to use them as reference manuals and look subjects up as necessary.

- *R for Data Science*, by H. Wickham, G. Grolemund. Available: <https://r4ds.had.co.nz/index.html>
- *The StatQuest Illustrated Guide to Machine Learning*, by Josh Starmer. Available on Amazon: \$27 for paperback. Based on videos available free at <https://www.youtube.com/@statquest>

- *Supervised Machine Learning for Text Analysis in R*, by E. Hvitfeldt and J. Silge. Available: <https://smldtar.com/>
- *Feature Engineering and Selection*, by M. Kuhn and K. Johnson. Available: <http://www.feats.engineering/>
- *R Markdown Cookbook*, by Y. Xie, C. Dervieux and E. Riederer. Available: <https://bookdown.org/yihui/rmarkdown-cookbook/>
- *Tidy Modeling with R*, by M. Kuhn and J. Silge. Available: <https://www.tmwr.org/>

Assessments

YOUR ATTAINMENT of course learning outcomes will be measured by the following assessments, with the relative weighting indicated in parentheses. All assignments within each category are given equal weight.

Collaboration and open science are fundamental aspects of data science. You can (and in fact are encouraged) to use the Internet, StackOverflow, etc. as a resource when coding. However, you **should not** directly copy code or answers. Your code should be well documented; include comments describing each line. If you use a source for help coding, you **must** cite it. You **are permitted** to collaborate/work in groups on the homework if you choose; if you choose to do so, you should list the names of the students you worked with at the beginning of your submitted homework assignment. **Everyone must complete their own final project; it is not a group assignment.**

You are permitted to make use of multimodal large language models when writing the **code** portions of assignments if you choose. However, you should be aware that the work of such models is often flawed at best, if not outright inaccurate, and be prepared to double- and triple-check your work before submission. If you do use an LLM for **any** part of your code, you absolutely **must cite** the use of that LLM, specifically indicating which tool you used (ChatGPT, Claude, DeepSeek, etc.) and which lines it was used for. Should we detect your use of any AI **without** such a citation, it will be considered plagiarism and reported to the Office of Student Conduct. You are **not permitted** to use any AI to write the text portions of assignments; should we detect such use, it will also be considered plagiarism. This restriction exists because written explanations are used to assess your own conceptual understanding, reasoning, and ability to communicate statistical ideas. (**Editing, paraphrasing, or rewording** text is also considered writing.)

Although the materials for the course are written using R, there may be students who prefer to complete the work using Python instead. You are completely free to use Python as an alternative to R (and Jupyter notebooks as an alternative to RStudio) if you choose. However, if you use Python for this course, to receive full credit for assignments you must **complete the exact tasks required**; in other words, you must complete every part of the assignment using the equivalent libraries and functions in Python. You may also switch back and forth, using Python for one assignment and R for another, if you choose. Note that course staff may be better versed in one language or the other.

- **Homework** (30%). Homework is assigned approximately every two weeks (this may vary) and will always be due by Sundays at 11:59 PM PDT. It will be **graded for accuracy** and solutions for the homework assignments will be posted approximately one week after each deadline. The graduate students will need to complete additional questions, but all assignments are ultimately rescaled to be out of 50 points. There are a total of **five** homework assignments and none will be dropped.
- **Labs** (20%). Section attendance is mandatory, and will be measured through the completion of a lab. Upon participating and completing the labs, you will receive credit for section attendance. You may miss a total of **one** section without penalty. Doing the labs will be greatly beneficial for your homework and project grades; with that in mind, you are strongly encouraged to attend all lab sections and follow along with the materials for each lab.
- **Final Project** (50%). The final project is a significant portion of the class and allows students to demonstrate their understanding of the material by working with a data set of their own choosing. Each student should begin thinking about their project and looking for a data set immediately; a page

on Canvas provides a list of possible resources. You will receive a considerable amount of guidance about the project over the course of the quarter.

The project will require a large amount of work and cannot be satisfactorily completed in only a day (or even a week). You are **strongly advised** to work on it in stages throughout the quarter. See Canvas for more detailed instructions, a rubric, and several examples of exemplary past projects.

The project will be graded not only on the code and model fitting, but also on the quality of the analysis process, justification of modeling choices, and clarity of written communication.

Course Schedule

THE WEEKLY schedule is indicated below. The topics and reading are subject to change based on the progress of the class.

	Topic	Assignment	Project Stage
Week 1	Intro to ML	Data Memo	Pick topic
Week 2	Bias-variance tradeoff	Homework 1	Find data
Week 3	Linear models		Tidy data
Week 4	Classification models	Homework 2	
Week 5	Discriminant analysis		
Week 6	Resampling & feature engineering	Homework 3	EDA
Week 7	Regularization		
Week 8	Tree-based methods	Homework 4	Run models
Week 9	Unsupervised learning		Write-up
Week 10	SVMs	Homework 5	Rough draft
Finals Week		Final project	Final draft

Time Commitment

THE COURSE is 4 credit units; each credit unit corresponds to an approximate time commitment of 3 hours. You should expect to allocate 12 hours per week to the course. If you find yourself spending considerably more time on the course on a regular basis, please let the instructor or TAs know so that we can help you balance the workload.

A suggested allocation of this time is as follows:

- Reading and class time: 3 hours (25%)
- Homework: 4.5 hours (37.5%)
- Sections: 1 hour (8.3%)
- Final project: 4.5 hours (37.5%)

Course Policies

Communication

THERE ARE FOUR means of communication with other students or the instructional team: during/after class, office hours, email, and individual appointments. **Please use them in that order.**

1. **During/after class.** The easiest guaranteed way to contact me is to ask a question before or after class, or raise your hand and ask a question during class. Note that any questions asked **during** class should be relevant to the material being discussed. Before or after class, you are free to ask any questions you may have.

2. **Office hours.** Office hours are offered at a minimum of twice weekly. I hold one session of 2 hours a week, and the other(s) are held by various combinations of TAs. You can attend any office hour session(s) that work with your schedule. These are valuable opportunities to interact informally, ask questions, and discuss course material or assignments.
3. **Email.** Please use email with discernment for simple communication. A response is guaranteed within 48 weekday hours (so if you email on Friday afternoon, you may not receive a reply until Tuesday afternoon). If your message is time-sensitive, please indicate so in the subject, and I will do my best to respond promptly. Emails asking questions that have already been answered in the syllabus or on Canvas may be briefly redirected without a detailed response.
4. **Appointment.** You can schedule individual 20-minute appointments with me as needed. These appointments may be either on Zoom or in person. This mode of communication is best suited to more complex or nuanced communication regarding personal matters. If you schedule an appointment, you will be prompted to indicate what you wish to discuss.

Grades

YOUR OVERALL GRADE in the course will be calculated as the weighted average of the proportions of total possible points in each assessment category according to the weightings indicated in the Assessments section and reported as a percentage rounded to two decimal places. Letter grades will be assigned according to the rubric below.

A+	100%
A	94% – 99.99%
A-	90% – 93.99%
B+	87% – 89.99%
B	84% – 86.99%
B-	80% – 83.99%
C+	77% – 79.99%
C	74% – 76.99%
C-	70% – 73.99%
D+	67% – 69.99%
D	64% – 66.99%
D-	60% – 63.99%
F	0% – 59.99%

Final course grades are assigned strictly according to this scale; grades are not rounded beyond the stated cutoffs.

YOU CAN keep track of your marks on individual assessments, your marks in each assessment category, and your overall grade in the Canvas gradebook. Please notify the instructor or TAs of any errors in grade *entry*; please do not attempt to negotiate the grades themselves. If at the end of the course you believe your grade was unfairly assigned, you are entitled to contest it according to the procedure outlined here in the UCSB General Catalog.

Deadlines

YOU RECEIVE two free late homework submissions without penalty. This policy applies only to homeworks. When you wish to use one of these late submissions, simply submit within one week of the original deadline, and **add a note** at the beginning of the assignment explaining that it is a free late submission.

NON-EXEMPTED homeworks submitted within one week of the deadline will be awarded 50% credit. **No credit** will be awarded for homework more than one week late; please plan ahead and submit your work on time.

THE FINAL PROJECT deadline is firm and no late submissions will be accepted.

Extensions

EXTENSIONS may be granted based on individual circumstances at the instructor's discretion. Extensions will never be granted retroactively and must be requested prior to the due date for an assignment.

Conduct

PLEASE BE MINDFUL of maintaining respectful and kind communication. You are expected to uphold the UCSB student code of conduct in your behavior when in class, in section, or interacting with other students or the instructional team. You can find the student code of conduct on the Office of Student Conduct website from this page. If you are uncomfortable with the conduct of another individual for any reason, please notify the instructor or TAs.

Academic Integrity

PLEASE MAINTAIN INTEGRITY in learning. Your work in the course must be your own. Any form of plagiarism, cheating, misrepresentation of individual effort on assignments and assessments, falsification of information or documents, or misuse of course materials compromises your own learning experience, that of your peers, and undermines the integrity of the UCSB community. Any evidence of dishonest conduct will be discussed with the student(s) involved and reported to the Office of Student Conduct. Depending on the nature of the evidence and the violation, penalty in the course may range from loss of credit to automatic failure. For a definition and examples of dishonesty, a discussion of what constitutes an appropriate response from faculty, and an explanation of the reporting and investigation process, see the OSC page on academic integrity.

Accommodations

REASONABLE ACCOMMODATIONS will be made for any student with a qualifying disability. Such requests should be made through the Disabled Students Program (DSP). More information, instructions on how to access accommodations, and information on related resources can be found on DSP website. Remote learning may present unique accommodation needs requiring additional flexibility; students receiving accommodation via DSP are invited to discuss this with the instructor if desired.

Student Evaluations

TOWARD THE END of the term, you will be given an opportunity to provide feedback about the course. Your suggestions and assessments are essential to improving the course, so please take the time to fill out the evaluations thoughtfully.

Student Resources

ANY STUDENTS in need are encouraged to make use of the following resources.

- Financial Crisis Response Team
<https://food.ucsb.edu/about/committees/financial-crisis-response-team>
- Food Security and Basic Needs (Food, Housing, Technology) Advising Center
<https://food.ucsb.edu/resources/basic-needs-advocates>
- Undocumented Student Services
<http://www.sa.ucsb.edu/dreamscholars/home>
- Campus Advocacy, Resources, and Education (CARE)
<https://care.ucsb.edu/>
24/7 Confidential Phone: (805) 893-4613
- The Trevor Project
<https://www.thetrevorproject.org/>

- Counseling and Psychological Services (CAPS)
<https://caps.sa.ucsb.edu/>
24/7 Counselors: (805) 893-4411, press 2